

Assignment#02 Cyber Security

LEGACY PROTOCOL INSECURITY & MODERN MITIGATION



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Strategic Analysis & Solutions

Legacy Protocol Insecurity & Modern Mitigation

Core Message: The evolution of network protocols mandates cryptographic security.

Introduction & Context

- Examines function and failure of Telnet protocol.
- Addresses architectural vulnerabilities allowing email spoofing.
- Early protocols prioritize ubiquity over security in trusted environments.
- Modern Internet requires cryptographic integrity and sender authentication.

Key Topics

-  1. Defining Telnet and its Plaintext flaw.
-  2. How Telnet's Network Virtual Terminal (NVT) enables remote access.
-  3. The mechanism of email spoofing via SMTP exploitation.
-  4. Practical demonstration of raw protocol interaction (Telnet).
-  5. Mitigation: Enforcing email security using DMARC.

Telnet: The Insecurity of Plaintext Communication

Legacy Protocol Insecurity and Modern Mitigation



Definition & Context

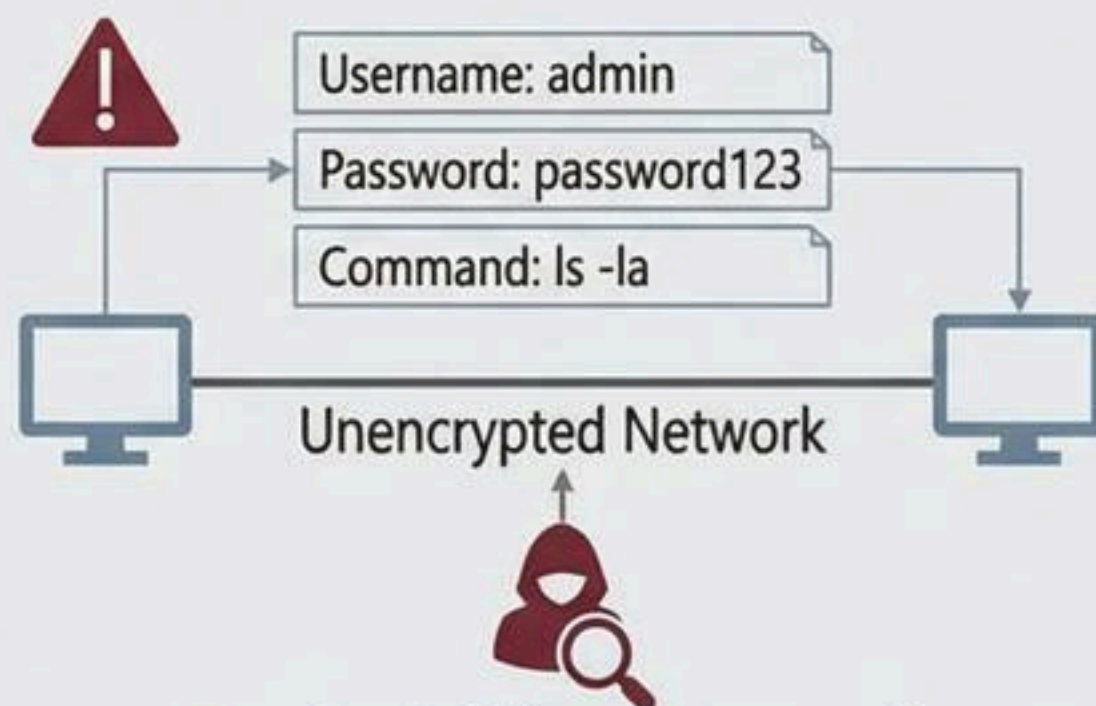
Teletype Network (Telnet): 1969 ARPANET protocol for bidirectional, text-based remote login.

Operates on TCP port 23 (Application Layer).

- Bridged early hardware incompatibilities.



The Critical Flaw: Plaintext Transmission



Passive Sniffing: Captures all keystrokes & credentials effortlessly.

Inherent design vulnerability: All data sent in unencrypted ASCII.



Consequence & Deprecation

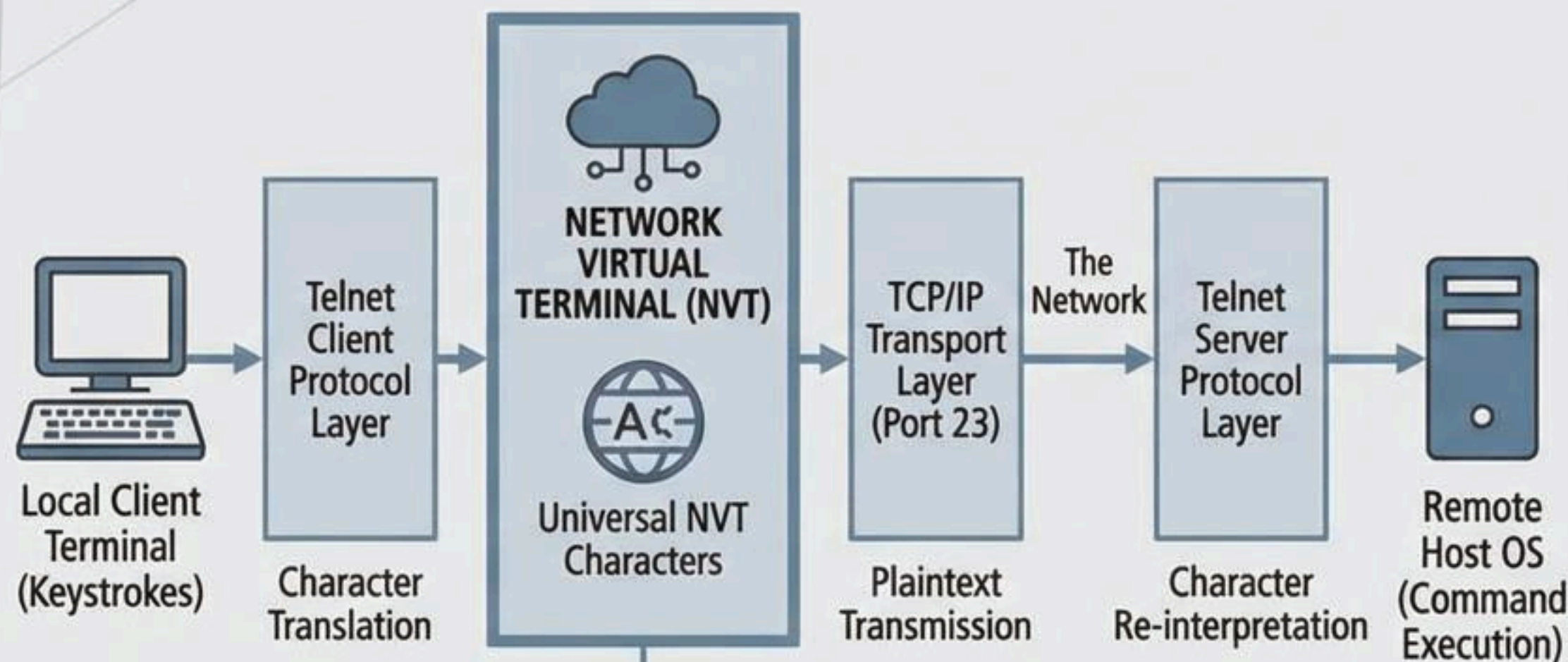


Critically Obsolete: Unacceptable for sensitive administration. Rendered obsolete by Secure Shell (SSH).

- SSH provides strong encryption (e.g., AES).
- Mandatory public-key authentication.
- Protects data integrity & confidentiality.

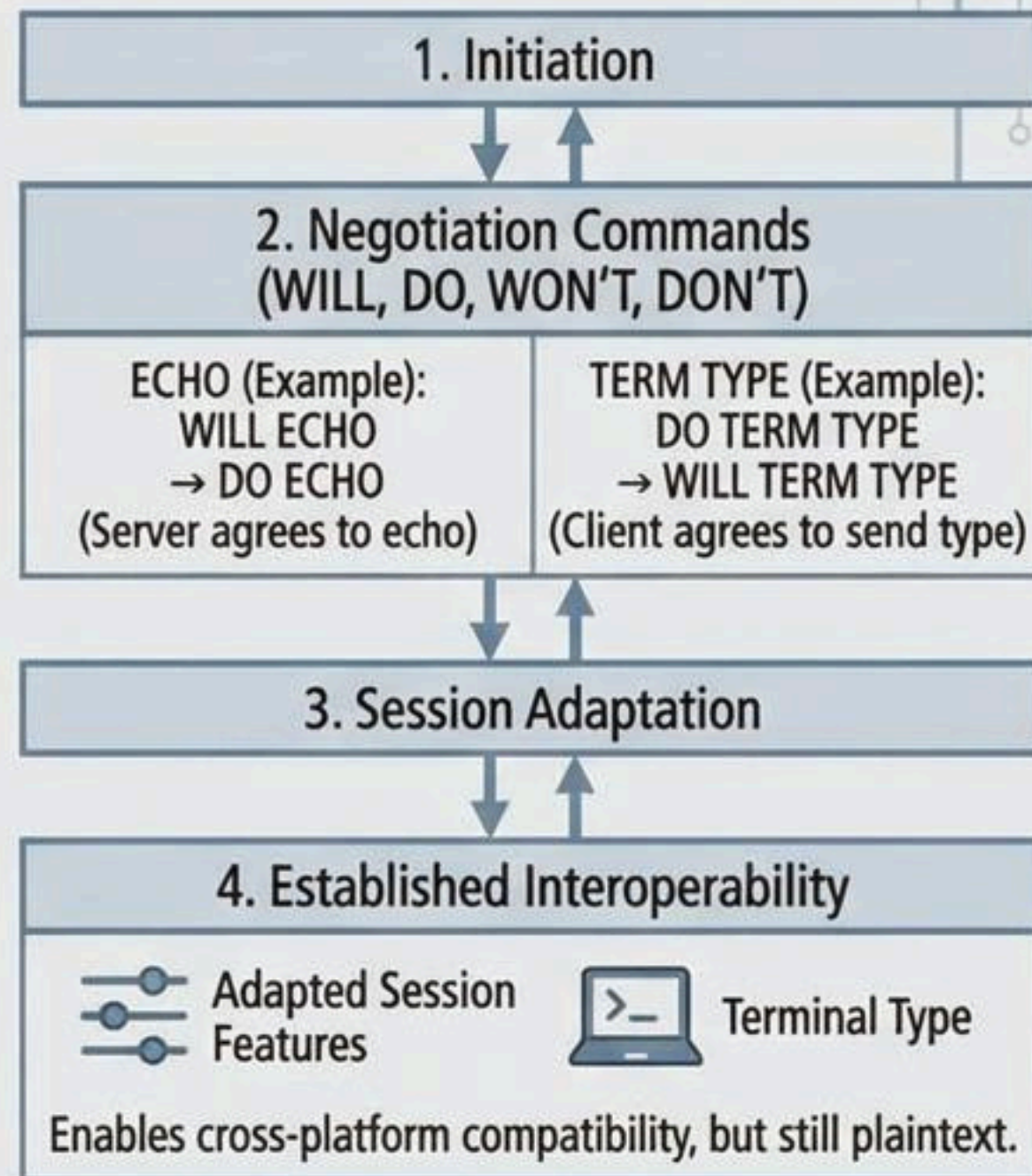
How Telnet Works: The Network Virtual Terminal (NVT) Model

The NVT Abstraction & Data Flow



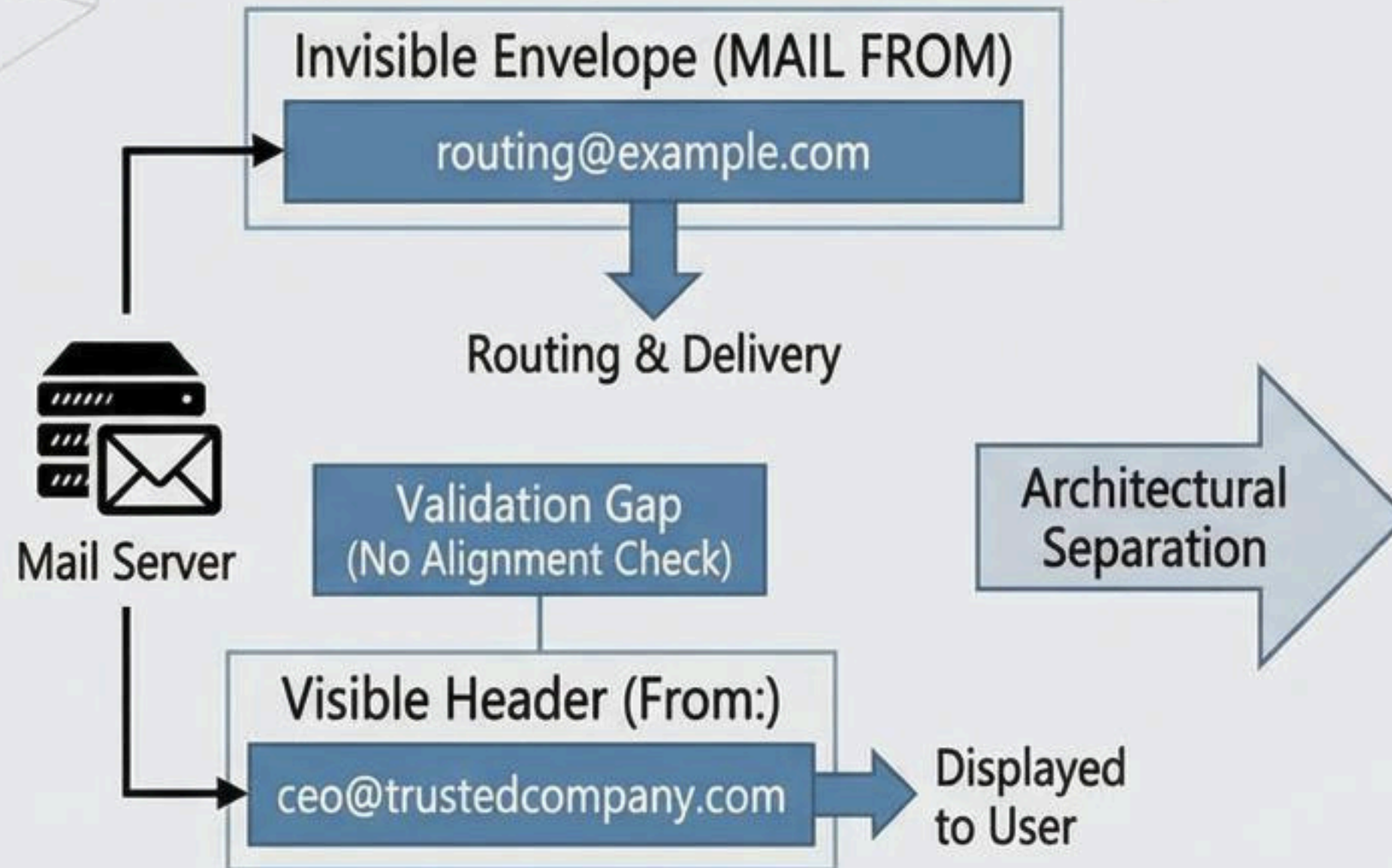
CRITICAL FLAW: All data (commands, credentials) travels in **UNENCRYPTED PLAINTEXT**.

Dynamic Option Negotiation

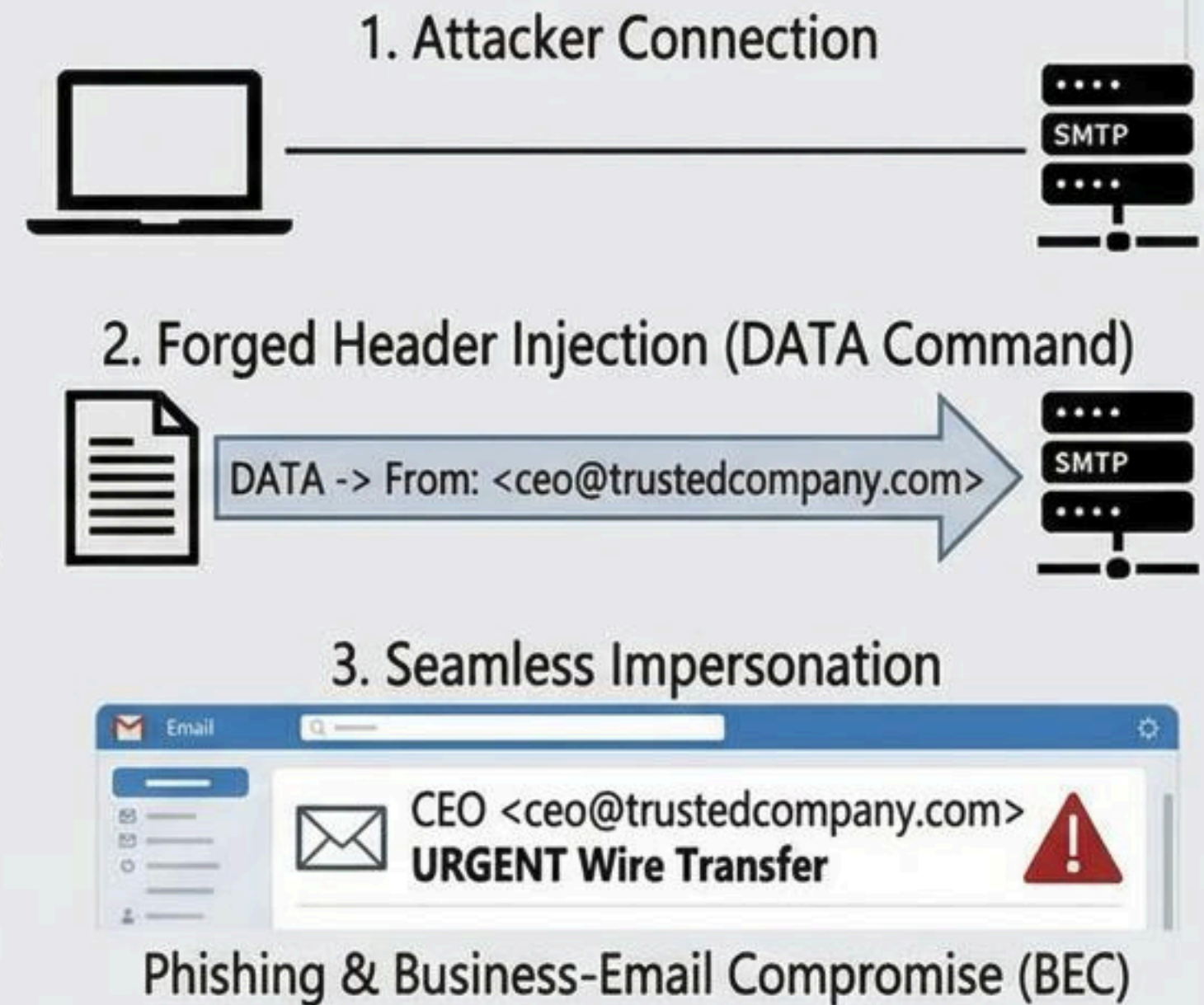


SMTP Protocol Exploitation: The Email Spoofing Mechanism

SMTP's Original Design (Routing vs. Display)



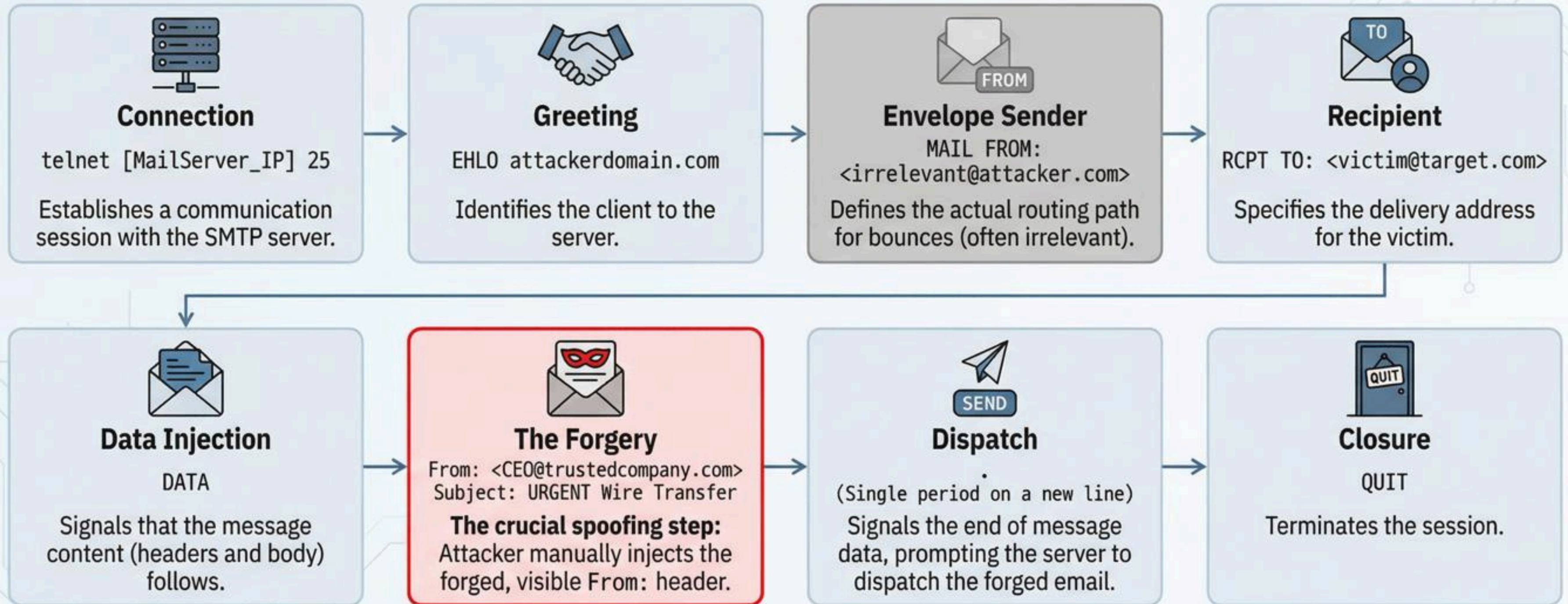
The Attack: Exploiting the Gap (Impersonation)



This structural vulnerability in SMTP allows attackers to inject forged visible sender information, enabling widespread phishing and security breaches.

Practical Example: Manual SMTP Session Forgery

Demonstrates the core technical manipulation required for spoofing via raw protocol interaction (Telnet to TCP port 25).



Proves that raw SMTP accepts unaligned headers without verification, completing the forgery in eight simple commands.

Legacy Protocol Insecurity & Modern Mitigation

Core Message: The evolution of network protocols mandates cryptographic security.

Modern Email Defense Layers



SPF (Sender Policy Framework) & **DKIM** (DomainKeys Identified Mail) - DNS Records



DMARC (Domain-based Message Authentication, Reporting, and Conformance) - Enforcement



Receiving Servers: Alignment Verification & Policy Action (Reject / Quarantine)

- Implements SPF, DKIM, and DMARC as interlinked DNS records.
- Receiving servers enforce alignment between the visible "From:" header and validated domains; failures trigger reject or quarantine policies.

Telnet's Safe Remaining Use: Banner Grabbing

```
> telnet google.com 80 [ENTER]
> HEAD / HTTP/1.1 [ENTER] [ENTER]
HTTP/1.1 200 OK...
Server: gws (Google Web Server)...
```



Safe raw protocol interaction to elicit application identifying information (the banner) via TCP port 80, illustrating service identification without exposing credentials.

Conclusions: Modern Security Posture

Core Message: The protocols of the early Internet are **inherently insecure** for the modern, untrusted network.

Legacy Protocol Weaknesses

Inherent Vulnerabilities & Failure



Telnet Security

- Fundamentally flawed; operates exclusively via plaintext transmission. Exposes login credentials and data to passive interception.



Email Security (SMTP)

- Widespread spoofing stems directly from structural failure to verify sender identity (header vs. envelope mismatch).

Modern Security Mandates

Layered Defense & Enforcement



SSH Mandate

- Secure Shell (SSH) is the non-negotiable industry standard for remote administration. Utilizes strong encryption and public-key authentication.



Defense-in-Depth (DMARC)

- Robust email security requires SPF and DKIM deployment, underpinned by DMARC. Enforces policy (reject/quarantine) and alignment to neutralize header forgery and protect against phishing.



Final Thought: Understanding legacy protocols provides foundational knowledge of network interaction, but contemporary practice dictates prioritizing layered security frameworks like SSH and DMARC to safeguard data integrity and system access.